

Student Name:



2021 Year 11 Examination

Mathematics Standard

Task 3: Examination

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks: 80

Section I – 15 marks (pages 2–6)

- Attempt Questions 1–15
- Write your answers on the answer sheet
- Allow about 25 minutes for this section

Section II – 65 marks (pages 7–21)

- Attempt Questions 16–32
- Allow about 1 hour and 35 minutes for this section

Disclaimer

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MSYR11_21_EXAM

Section I

15 marks

Attempt Questions 1–15

Allow about 25 minutes for this section

Use the multiple-choice answer sheet for Questions 1–15.

- 1 Express 1.0788 correct to 2 significant figures.
- A. 1.0
 - B. 1.07
 - C. 1.08
 - D. 1.1
- 2 What is the value of k in the equation $5(2k - 1) = 9k$?
- A. 1.4
 - B. 0.5
 - C. 2.5
 - D. 5
- 3 Calculate the percentage error for a measurement of 4.22 m, measured to the nearest centimeter. Round your answer correct to 2 decimal places.
- A. 0.12%
 - B. 0.24%
 - C. 11.85%
 - D. 23.70%
- 4 Freddy borrows \$17 000 at 3.5% p.a. flat rate interest to replace the roof on his house. How much interest does he owe after 9 months?
- A. \$37.19
 - B. \$5355
 - C. \$726
 - D. \$446.25

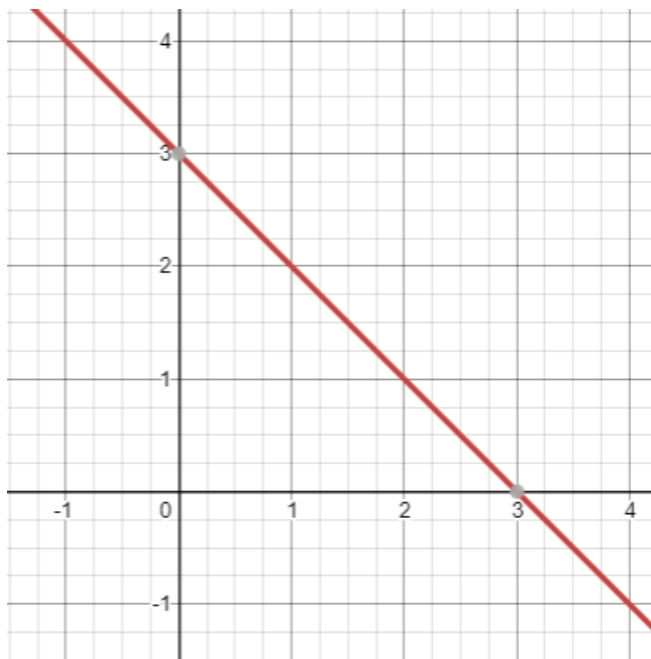
- 5 Fried's formula, shown below, is used to calculate the required dosage of medicine for children aged 1 – 2 years.

$$\text{Dosage} = \frac{\text{Age of infant (months)} \times \text{Adult dose}}{150}$$

Calculate the dosage for an infant aged 1.5 years when the corresponding adult dose is 45 mg.

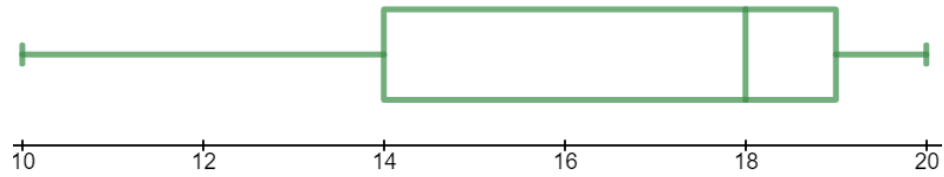
- A. 0.45 mg
- B. 5.4 mg
- C. 0.375 g
- D. 4.5 g

- 6 What is the equation of the line shown?



- A. $y = 3 - x$
- B. $y = 3 + x$
- C. $y = 3x + 3$
- D. $y = 3 - 3x$

- 7 Which term correctly describes the shape of the data displayed?



- A. Negatively skewed
B. Positively skewed
C. Symmetrical
D. Uniform
- 8 It takes David 23 minutes to travel to school. If school starts at 8.40 am, what is the latest time he can leave home to arrive on time?
- A. 8.17 am
B. 8.23 am
C. 8.27 am
D. 9.03 am
- 9 Georgia earned \$1032 in a week where she worked 38 hours at normal pay, 4 hours at time and a half and 2 hours at double time. Calculate her hourly rate of pay.
- A. \$11.30
B. \$21.50
C. \$23.45
D. \$28.60
- 10 A three-digit number is formed from the digits 0, 3, 4, 7, 9 with no repetition. What is the probability the number will be odd?
- A. 0.3
B. 0.4
C. 0.6
D. 0.8

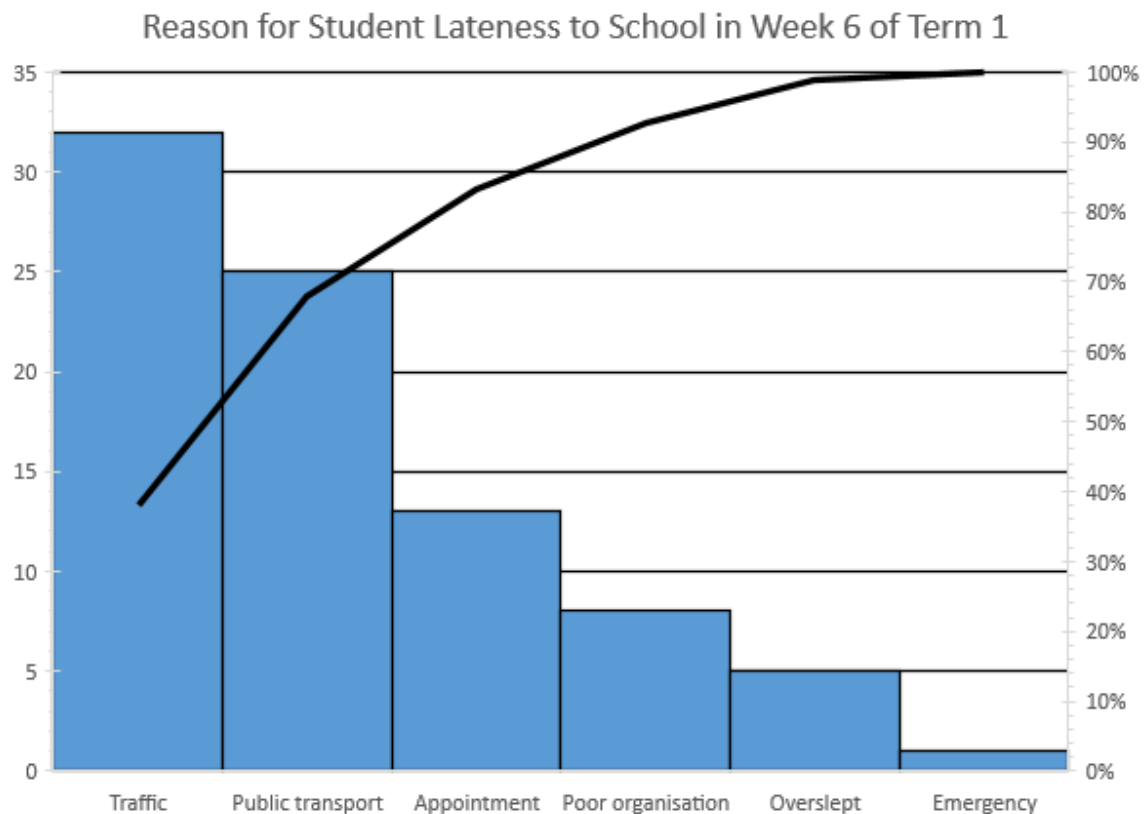
11 What is the time in Fiji (UTC+12) if it is 3.00 am in Paris (UTC+2)?

- A. 5 am
- B. 1 pm
- C. 3 pm
- D. 5 pm

12 d varies directly with t . What is the constant of variation when $d = 135$ and $t = 4.5$?

- A. 607.5
- B. 139.5
- C. 131.5
- D. 30

13 From the following Pareto chart, determine the percentage of students who were late to school due to an appointment. Round to the nearest percent.



- A. 13%
- B. 15%
- C. 37%
- D. 83%

- 14** The cost of a pair of shoes is \$129 including GST.
What was the cost before GST was added (rounded to the nearest cent)?
- A. \$116.10
 - B. \$117.27
 - C. \$119
 - D. \$141.90
- 15** The coordinates of Swakopmund, Namibia are 22.6°S, 14.6°E. Mayo-Rey, Cameroon is 30 degrees North of Swakopmund.
What are the coordinates of Mayo-Rey?
- A. 52.6°S, 14.6°E
 - B. 52.6°N, 14.6°E
 - C. 7.4°N, 14.6°E
 - D. 7.4°S, 14.6°E

End of Section I

Mathematics Standard

Section II Answer Booklet

65 marks

Attempt Questions 16–32

Allow about 1 hour and 35 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Question 16 (1 mark)

The distance from Earth to the Moon is 384402 km. Convert this measurement to metres, expressing your answer in standard form. **1**

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Question 17 (1 mark)

Rearrange $v^2 = u^2 + 2as$ so that a is the subject of the formula. **1**

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Question 18 (2 marks)

Georgia earns an annual salary of \$84 000. When taking 4 weeks annual leave, she receives 17.5% of 4 weeks normal pay as leave loading in addition to her normal pay. Calculate her total annual leave for four weeks. **2**

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Question 19 (2 marks)

A data set of 5 numbers has a mode of 8, a median of 5, a range of 7 and a mean of 5. 2
List the numbers in this data set, showing all relevant working.

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Question 20 (4 marks)

The following back-to-back stem-and-leaf plot show the test results of two Mathematics classes.

Class A		Class B
	2	
7	3	9
9 5 3	4	0 1
	5	2 7
1	6	5 6 8 8 8 9
8 7 6 2 1 0	7	0 2 7 7 9
6 5 3	8	1 2 3 3 5 6
9 8 8 7 7 7 6 3 2	9	1 1 4
9 8		

(a) What was the highest mark in Class A? 1

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(b) What percentage of students in Class B scored less than 50? 1

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(c) Which class performed more consistently? Justify your answer with working. 2

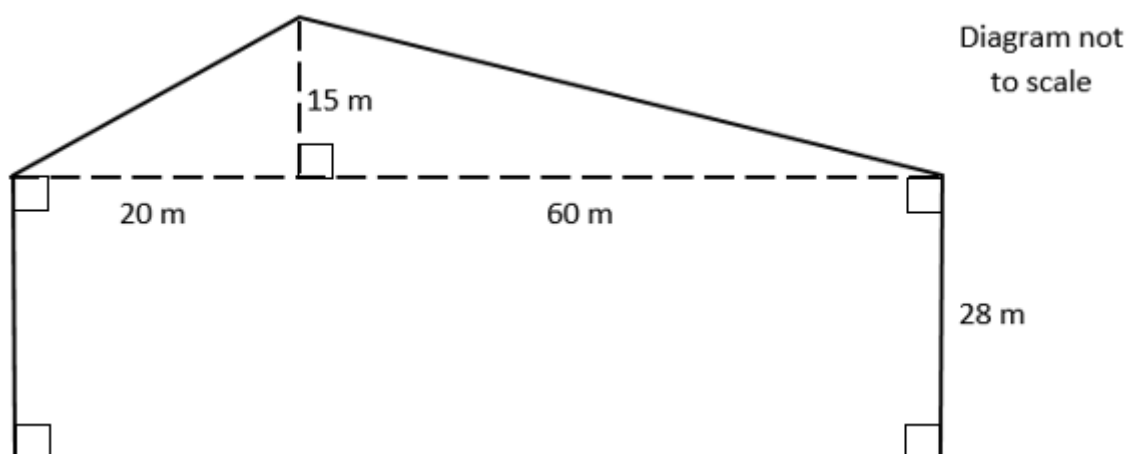
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Question 21 (6 marks)

The diagram below shows the top view of a park.



- (a) Calculate the perimeter of the park, to the nearest metre. 2

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- (b) The fencing chosen for this park costs \$54 per metre. Cost of labour to install the fence costs \$42 per metre. Calculate the total cost to fence this area. 2

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- (c) All measurements have been taken to the nearest metre. For the lower, rectangular section of the park, between which lower and upper limits does the actual area lie? 2

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Question 22 (8 marks)

The following are the 2 km race times (in minutes) for the schoolgirls' coxed eight rowing final.

6.88 7.15 6.93 7.07 7.30 7.10 6.95 6.98

- (a) Create a five-number summary for the above data. 3

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- (b) Calculate the standard deviation for the above data to 3 significant figures. 1

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- (c) Are there any outliers? Justify your answer with working. 2

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- (d) In general, state the effect of outliers on the mean and median in small data sets. 2

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Question 23 (3 marks)

Use the timetable below to answer the following questions.

Brisbane (Roma Street) – Byron Bay – Grafton – Moree – Sydney (Central)

			
Destination	Grafton	Sydney	Moree
Days of Operation	Daily	Daily*	Tue/Thu/Sat
Service No.	144	032	141
Brisbane (Roma St)		05:55	
Kyogle		a07:53	
Byron Bay	05:35	...	
Suffolk Park	05:40	...	
Lennox Head	05:55	...	
Ballina	06:10	...	
Ballina West	06:15	...	
Wardell	06:25	...	
Broadwater	06:30	...	
Evans Head	06:45	...	
Woodburn	06:55	...	
Iluka	07:35	...	
Woombah	07:45	...	
Chatsworth Island	07:52	...	
Yamba	08:15	...	
Yamba West	08:20	...	
Palmers Island	08:30	...	
Maclean	08:40	...	
Tyndale	08:50	...	
Cowper	08:55	...	
Ulmarra	09:05	...	
Casino	...	08:20	
Grafton	ARV 09:20		
Connection			
Grafton	DEP	09:53 → 10:20	
Jackadgery		...	10:55
Gibraltar Range		...	a11:35
Glen Innes		...	12:25
Inverell	ARV	...	13:15 
Inverell	DEP	...	14:15
Mount Russell turnoff		...	a14:25
Delungra		...	14:40
Warialda		...	15:00
Gravesend		...	a15:15
Biniguy		...	a15:30
Moree		...	16:05
Moree Town		...	16:10
Coffs Harbour		11:05	
Sawtell		a11:14	
Urunga		a11:31	
Nambucca Heads		11:47	
Macksville		12:00	
Eungai		...	
Kempsey		12:43	
Wauchope		13:22	
Kendall		a13:54	
Taree		14:41	
Wingham		14:55	
Gloucester		15:44	
Dungog		16:38	
Maitland		17:30	
Broadmeadow		17:52	
Fassifern		d18:11	
Wyong		d18:42	
Gosford		d18:57	
Hornsby		d19:38	
Strathfield		d19:58	
Sydney (Central)		20:12	

Explanation of definitions and symbols

a	Stops to pick up and set down booked customers only when required
b	Stops to set down booked customers only when required
c	Stops to pick up booked customers only when required
d	Stops to set down booked customers only
u	Stops to pick up booked customers only
*	Daylight Savings NSW. During this time services arrive or depart Queensland locations one (1) hour earlier than shown (QLD time).
	Refreshment break – usually 30 minutes. Length of break may change when required.
	Shows train/coach connection
	NSW TrainLink train service
	NSW TrainLink coach service
	NSW TrainLink Taxi service
	Wheelchair Accessible

* During NSW Daylight Savings, services will arrive and depart QLD locations 1 hour earlier than shown.

Source: [Transport NSW](https://www.transport.nsw.gov.au)

Question 23 continues on next page

Question 23 (continued)

- (a) How long will it take Mateo to travel from Taree to Strathfield? **1**

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- (b) What time does the train from Brisbane to Sydney depart Grafton? **1**

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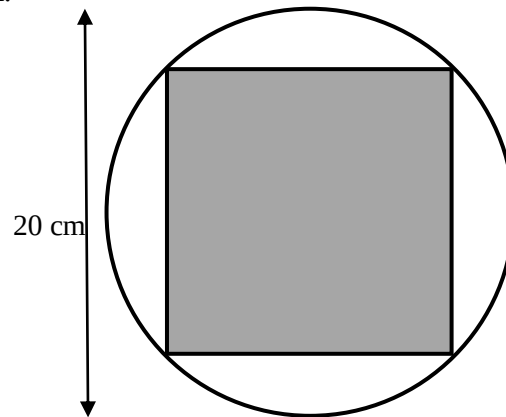
- (c) What time does David need to leave Brisbane, QLD, to arrive in Coffs Harbour NSW at 11:05 during NSW Daylight Savings? **1**

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End of Question 23

Question 24 (2 marks)

A cake shop has run out of square cake boxes, so they are having to use their round boxes for all their cakes. The following is the top view of a square cake that only just fits into a cylindrical cake box.



Calculate the height of the box needed if the diameter of the cylindrical box is 20 cm and the volume of the cake is 2400cm^3 . 2

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Question 25 (3 marks)

A plasma television uses 140 watts per hour and is used an average of 4 hours per day.

- (a) Calculate the cost of electricity used by the television over a year at \$0.12 per kWh. 2

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- (b) A family is considering purchasing a LED television which uses 90 watts per hour. What will be the percentage of energy saved, correct to 1 decimal place? 1

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Question 26 (8 marks)

Jasmine purchased a new car for \$22 000.

- (a) Her car depreciates by \$850 each year, what is its value after 3 years? **1**

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- (b) What is the percentage decrease in the price over the 3 years correct to 2 decimal places? **1**

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- (c) Jasmine paid 10% deposit and took out a loan for the remainder of the cost at 6.5% p.a. simple interest over 5 years. She repays the loan in equal monthly instalments. Show that her monthly repayment will be \$473.25. **4**

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- (d) Jasmine drives approximately 250 km per week and her car uses 7.2 L of petrol per 100 km. The average cost of petrol is \$1.42 per litre. How much does she pay for petrol in one year? **2**

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Question 27 (2 marks)

A bag contains yellow, blue and green marbles. The probability of selecting a yellow marble is 22%, the probability of selecting a blue marble is $\frac{5}{8}$. A marble is selected and replaced 400 times. What is the expected number of times a green marble would be drawn?

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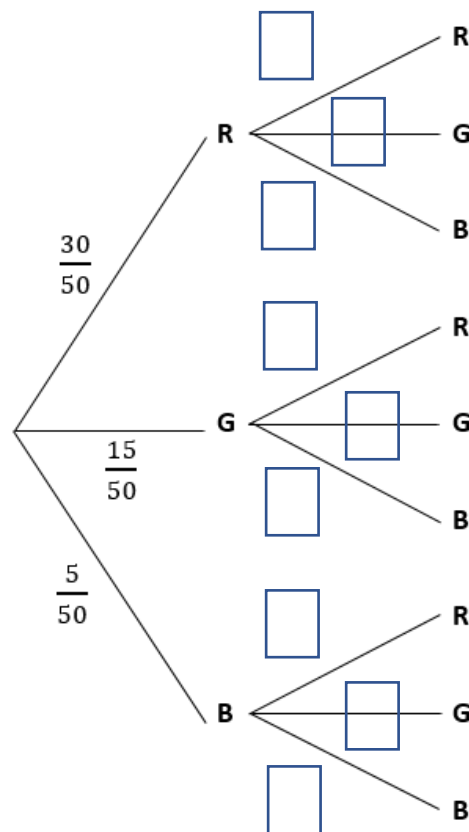
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Question 28 (4 marks)

A jar contains 50 jellybeans. 30 are red, 15 are green and 5 are black. Two jellybeans are selected at random without replacement.

(a) Complete the following probability diagram by adding probabilities to each branch.

2



(b) What is the probability of selecting two jellybeans that are the same colour?

2

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Question 29 (5 marks)

Use the tax table below to answer the following question.

Resident tax rates 2020-21

Taxable income	Tax on this income
0 - \$18,200	Nil
\$18,201 - \$45,000	19 cents for each \$1 over \$18,200
\$45,001 - \$120,000	\$5,092 plus 32.5 cents for each \$1 over \$45,000
\$120,001 - \$180,000	\$29,467 plus 37 cents for each \$1 over \$120,000
\$180,001 and over	\$51,667 plus 45 cents for each \$1 over \$180,000

The above rates **do not** include the Medicare levy of 2%.

Source: ATO

Andrea's fortnightly gross income is \$2150. She pays \$352 PAYG tax each fortnight and has annual deductions totalling \$550. She must also pay the Medicare levy which is calculated at 2% of her taxable income. Determine whether Andrea will receive a tax refund or if she will have to pay more. Justify your answer with working.

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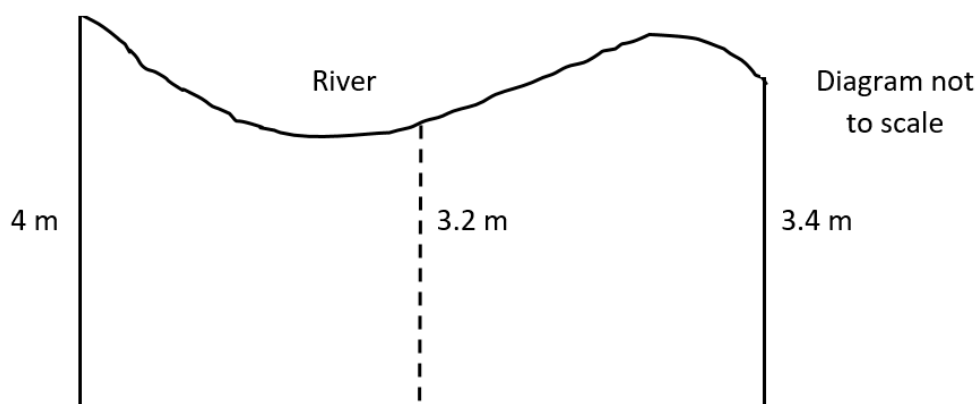
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Question 30 (4 marks)

A campsite borders a river and is 5.4 m wide. Other measurements are shown on the diagram below.



- (a) Use two applications of the trapezoidal rule to calculate the area of the campsite. 2

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- (b) Occasionally in heavy rain this campsite floods. What is the volume of water, in litres, covering this campsite if the water rises to 4 cm above ground level? 2

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Question 31 (5 marks)

A family have gone on a 2-week holiday leaving their kitchen tap dripping at a constant rate. The sink the tap drips into has a capacity of 50 L and the plug has been left in. After 10 hours of water dripping from the tap, the sink contains 2 L of water. Let W represent the amount of water in the sink at any time, t hours.

- (a) Calculate the constant of variation. What does it represent? 2

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- (b) Write the equation of the line that models this relationship after the tap started dripping. 1

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- (c) Will the family return from their holiday before their sink overflows? 2
Justify your answer with working.

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Question 32 (5 marks)

Laura and Sally have been to their cousin's engagement party, they have been drinking but hope to drive home. Laura weighs 62 kg and has consumed 4 standard drinks over 4 hours. Sally weighs 75 kg and has consumed 6 standard drinks over 4 hours. Laura has a provisional license (P plates) for which the blood alcohol concentration (BAC) limit is 0. Sally has a full licence for which the blood alcohol concentration (BAC) limit is 0.05. What is the earliest time, to the nearest minute, that either of them could drive home if they stopped drinking at 10 pm? 5

$$BAC_{Female} = \frac{(10N - 7.5H)}{5.5M}$$

$$\text{Number of hours for BAC to reach zero} = \frac{BAC}{0.015}$$

BAC: Blood alcohol content

N: Number of standard drinks consumed

H: Hours drinking

M: Mass in kilograms

Number of hours: Number of hours for BAC to reach zero

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

Mathematics Standard Reference Sheet

Measurement

Precision

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2} (x + y)$$

$$A = \frac{h}{2} (d_f + d_l)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

$$z = \frac{x - \bar{x}}{s}$$

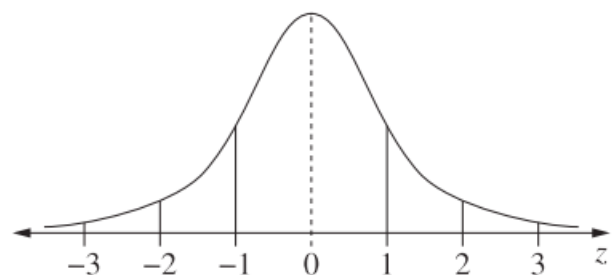
An outlier is a score

$$\text{less than } Q_1 - 1.5 \times IQR$$

or

$$\text{more than } Q_3 + 1.5 \times IQR$$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3